AI Image Generator

7. Database

Version 1.0
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Platform: NVIDIA GeForce RTX 3090, Windows 11, Python 3.12

Stable Diffusion XL + Real-ESRGAN, served locally with Gradio, with per-image GPU telemetry recorded in SQLite.

7. Database

7.1 Overview

File: generations.db in the project folder. Engine: SQLite 3 through Python's standard library. One table, generations, one row per image produced by the web app. An index on created_at supports newest-first listing.

The schema is created on first connection. Columns added in later increments are listed in NEW_COLUMNS and added with ALTER TABLE when absent, so an older database file is upgraded in place the first time the new code opens it.

7.2 Table: generations

Identity and files

Column	Type	Meaning
id	INTEGER PK	Auto-increment row id, shown as "#N" in the UI
created_at	TEXT	ISO-8601 local timestamp, seconds precision
image_path	TEXT	Absolute path of the native PNG
file_size_kb	REAL	Size of the native PNG

Prompt

Column	Type	Meaning
user_prompt	TEXT	Exactly what was typed
final_prompt	TEXT	What was sent to the model after directive removal and style suffixes
negative_prompt	TEXT	Negative prompt used
parser_notes	TEXT	Human-readable list of parser decisions, joined with " - "

Settings

Column	Type	Meaning
model_id	TEXT	Hugging Face model id
width, height	INTEGER	Native generation size
steps	INTEGER	Denoising steps
guidance	REAL	Classifier-free guidance scale
seed	INTEGER	Seed used for this image
style	TEXT	Name of the matched style preset, or NULL
batch_index, batch_size	INTEGER	Position within a multi-image request

Timing

Column	Type	Meaning
inference_seconds	REAL	Pipeline call, from start to CUDA synchronise
save_seconds	REAL	PNG encode and write of the native image
total_seconds	REAL	inference + save (excludes upscaling)
steps_per_second	REAL	steps / inference_seconds

GPU and software

Column	Type	Meaning
gpu_name	TEXT	From torch device properties
vram_total_gb	REAL	Card memory
vram_peak_gb	REAL	Peak torch allocation during this image, including upscaling
vram_reserved_gb	REAL	Allocator reservation after the image
compute_capability	TEXT	e.g. "8.6"
driver_version	TEXT	From nvidia-smi
cuda_version, torch_version, diffusers_version	TEXT	Library versions

Telemetry (from nvidia-smi)

Column	Type	Meaning
temp_before_c, temp_after_c	REAL	GPU temperature before the pipeline call and right after it
power_before_w, power_after_w	REAL	Board power draw at the same two moments
power_limit_w	REAL	Configured power limit
gpu_util_after_pct	REAL	GPU utilisation at the after snapshot
sm_clock_mhz, mem_clock_mhz	REAL	Clock speeds at the after snapshot
fan_after_pct	REAL	Fan speed
pstate	TEXT	Performance state, P0 to P8

Upscaling (added in increment 5)

Column	Type	Meaning
upscale_target	INTEGER	Requested long side in pixels
upscaled_width, upscaled_height	INTEGER	Final size after resample and crop
upscaled_path	TEXT	Absolute path of the upscaled PNG
upscaled_size_kb	REAL	Size of that file
upscale_seconds	REAL	Time of the upscaling stage
upscale_passes	INTEGER	Number of x4 passes (1 or 2)
upscaler	TEXT	Architecture and weight file name

Catch-all

Column	Type	Meaning
extra_json	TEXT	JSON of any record keys that have no column, for forward compatibility

7.3 Example queries

Average inference time by native size:

```
SELECT width || 'x' || height AS size, COUNT(*) AS n,
       ROUND(AVG(inference_seconds), 2) AS avg_s
FROM generations GROUP BY size ORDER BY n DESC;
```

Hottest runs:

```
SELECT id, temp_after_c, power_after_w, user_prompt
FROM generations ORDER BY temp_after_c DESC LIMIT 5;
```

Every 4K or larger image with its file:

```
SELECT id, upscaled_width, upscaled_height, upscale_seconds, upscaled_path
FROM generations WHERE upscale_target >= 3840;
```

7.4 Access from Python

```
import db
rows = db.fetch(limit=10, search="cabin")
one = db.get(4)
totals = db.stats()
```